PrecisionPlus v3™ Report

Patent Pending

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Patent Pending

MRI OF THE LUMBAR SPINE

CLINICAL INDICATION: Low back pain following slip and fall on 02/11/2026.

TECHNIQUE: T1 and T2 weighted images were obtained.

COMPARISON: No prior studies are available for comparison.

FINDINGS:

Unless otherwise specified in this report, the lumbar level numbering is based upon the most 5 caudad well-formed vertebral bodies, being numbered L1 through L5. Prior to any possible surgical planning or any possible instrumentation, exact confirmation is required to determine the appropriate level.

The conus medullaris appears normal. The lordotic curvature of the lumbar spine is preserved. No evidence for abnormal solid or cystic lesions is identified. No prevertebral or paravertebral masses or fluid collections are seen and there is no evidence for abnormal marrow replacing lesion.

Segmental analysis of the lumbar spine is as follows:

At T12-L1, there is a central disc herniation superimposed on a disc bulge, the combination of which measures 2.5 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. There are anterior and there are posterior osteophytes, however the disc material extends posterior to the posterior osteophytes. There is no spinal stenosis or neural foraminal stenosis.

At L1-2, there is a disc bulge measuring 2.5 mm superimposed on a 2 mm grade 1 retrolisthesis of L1 on L2 indenting the ventral thecal sac and elevating the posterior longitudinal ligament. There is no spinal stenosis or neural foraminal stenosis. There are anterior osteophytes and posterior osteophytes seen. There is facet arthropathy.

At L2-3, there is a central disc herniation superimposed on a disc bulge, the combination of which measures 2.5 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. There are anterior and there are posterior osteophytes, however the disc material extends posterior to the posterior osteophytes. There is no spinal stenosis or neural foraminal stenosis.

At L3-4, there is a right neural foraminal disc herniation superimposed on a disc bulge, the combination of which measures 3 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. This is superimposed on a 2 mm grade 1 retrolisthesis of L3 on L4. There is no spinal stenosis or neural foraminal stenosis. There is facet arthropathy.

At L4-5, there is a left neural foraminal disc herniation measuring 3.5 mm indenting the ventral thecal sac and elevating the posterior longitudinal ligament. There is a zone of hyperintensity within the disc consistent with an annular fissure. There are anterior and there are posterior osteophytes, however the disc material extends posterior to the posterior osteophytes. There is mild left neural foraminal narrowing. There is no right neural foraminal narrowing or spinal stenosis. There is facet arthropathy with a mild right facet joint effusion.

At L5-S1, there is a central disc herniation superimposed on a disc bulge, the combination of which measures 4 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. There is moderate right neural foraminal narrowing. There is no left neural foraminal narrowing or spinal stenosis.

There is mild spinal asymmetry with apex right at L4.

There is decreased intervertebral disc signal at T12-L1 through L5-S1.

There is decreased intervertebral disc space height at L1-2.

There is a hemangioma in T12, L2 and L5.

There are Schmorl's nodes at multiple endplates in the lower thoracic and lumbar spine.

There is mild anterior wedging of the L1 vertebral body, chronic appearing.

There is increased signal within the interspinous ligament at L5-S1 consistent with interspinous ligament injury/edema.

Enlarged prostate gland. Correlate with PSA.

IMPRESSION:

- At T12-L1, there is a central disc herniation superimposed on a disc bulge, the combination of which measures 2.5 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. There are anterior and there are posterior osteophytes, however the disc material extends posterior to the posterior osteophytes. *See **Figure 2** Sagittal T2 image #8.
- At L1-2, there is a disc bulge measuring 2.5 mm superimposed on a 2 mm grade 1 retrolisthesis of L1 on L2 indenting the ventral thecal sac and elevating the posterior longitudinal ligament.
- At L2-3, there is a central disc herniation superimposed on a disc bulge, the combination of which measures 2.5 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. There are anterior and there are posterior osteophytes, however the disc material extends posterior to the posterior osteophytes. *See **Figure 2** Sagittal T2 image #8.
- At L3-4, there is a right neural foraminal disc herniation superimposed on a disc bulge, the combination of which measures 3 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. This is superimposed on a 2 mm grade 1 retrolisthesis of L3 on L4.

- At L4-5, there is a left neural foraminal disc herniation measuring 3.5 mm indenting the ventral thecal sac and elevating the posterior longitudinal ligament. There is a zone of hyperintensity within the disc consistent with an annular fissure. There are anterior and there are posterior osteophytes, however the disc material extends posterior to the posterior osteophytes. There is mild left neural foraminal narrowing. There is facet arthropathy with a mild right facet joint effusion.
- At L5-S1, there is a central disc herniation superimposed on a disc bulge, the combination of which measures 4 mm. The ventral thecal sac is indented, and the posterior longitudinal ligament is elevated. There is a zone of hyperintensity within the disc consistent with an annular fissure. There is moderate right neural foraminal narrowing. *See **Figure 1** Sagittal T2 image #7.
- There is mild spinal asymmetry with apex right at L4.
- There is mild anterior wedging of the L1 vertebral body, chronic appearing.
- There is increased signal within the interspinous ligament at L5-S1 consistent with interspinous ligament injury/edema. *See **Figure 3** Sagittal STIR Image #8 arrow shows the increased signal consistent with interspinous ligament injury/edema.
- This MRI was performed for symptoms following an acute traumatic event. Given the patient's indication of symptoms following trauma, clinical correlation with the patient's treating physicians including symptomology following injury is recommended to determine and/or confirm that the above findings are related to the patient's trauma dated 02/11/2026. This diagnostic opinion should be used to inform the course and scope of treatment as clinically appropriate in accordance with the findings above.

Expert Radiology has initiated direct communication process with the referring physician's office via telephone.

It should be noted that facetogenic pain is a clinical diagnosis due to the irritation of the facet joints, and MRI is insensitive for symptomology such as facet-related pain. MRI Spine is insensitive for SI mediated/caused pain.
The definitions in this report, including definitions of disc bulge, disc herniation, protrusion, and extrusion, are from the following peer reviewed journal: Lumbar Disc Nomenclature V2.0, Recommendations of the Combined Task Forces of the North American Spine Society, the American Society of Spine Radiology and the American Society of Neuroradiology, The Spine Journal 14 (2014) 2525 2545. Disc herniations are protrusion type disc herniations unless specifically described as extrusions. A normal MRI should not supersede abnormalities detected with physical exam.**FIGURE 1:**

Image shows a central disc herniation at L5-S1 superimposed on a disc bulge measuring 4 mm, with indentation of the ventral thecal sac and elevation of the posterior longitudinal ligament, and a zone of hyperintensity within the disc consistent with an annular fissure. *(Automatic description of image based on radiologist's report)*

FIGURE 2:

Image shows a central disc herniation at T12-L1 superimposed on a disc bulge measuring 2.5 mm, with indentation of the ventral thecal sac and elevation of the posterior longitudinal ligament, and a zone of hyperintensity within the disc consistent with an annular fissure. *(Automatic description of image based on radiologist's report)*

FIGURE 3:

Image shows increased signal within the interspinous ligament at L5-S1 consistent with interspinous ligament injury/edema. *(Automatic description of image based on radiologist's report)*

COLORIZATION OF FIGURE 1:	**NORMAL LUMBAR SPINE:**
PATIENT LASTNAME, FIRSTNAME	*HEALTHY PATIENT*
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Note: The image on the LEFT is a colorized image of the patient's abnormality for illustrative purposes only, and this colorized image should not be used for medical decision-making

Thank you for the consultation. This MRI report was personally interpreted and finalized by the signing radiologist. To ensure the highest-quality MRI and CT reports with colorized key images and illustrations, please request **Expert Radiology v3™** on future order scripts.
Sincerely,

Scan to View this Radiologist's CV

Electronically Signed by HEWETT, LEE MD at 04/09/2026 12:26:17 PM UTC-05:00
Transcribed/ T.K
V3

Customized Spine Illustration of patient:
Sagittal View of Lumbar Spine
(Based upon Findings in the Report)

This illustration depicts a Sagittal view (side view) of the spine. It attempts to highlight some of the area(s) of interest as described in the radiology report. This illustration does not necessarily show all the pathology present. Please refer to the original radiologists' report. This illustration attempts to correspond to the level(s) of the pathology specified, providing a visual representation to improve the understanding of the reported findings for the patient. Protrusions and Extrusions are both subtypes of Herniation. The image selection was generated by a computer and has not been reviewed by the radiologist for accuracy, correctness, or completeness. The illustration should NOT be used in medical decision making or diagnostic purposes. The illustration is provided solely as a convenience for the patient and should not be relied upon as a medical diagnosis. If you have questions relating to this illustration please contact your personal physician. The illustration is not intended to be a medical device and is not subject to FDA oversight.

Customized Spine Illustration of patient:
Axial View of Lumbar Spine Level L5-S1
(Based upon Findings in the Report)

This illustration depicts an Axial view (top down view) of the spine. It attempts to highlight some of the area(s) of interest as described in the radiology report. This illustration does not necessarily show all the pathology present. Please refer to the original radiologists' report. This illustration attempts to correspond to at least one of the position(s) of the pathology specified, providing a visual representation to

improve the understanding of the reported findings for the patient. Protrusions and Extrusions are both subtypes of Herniation. The image selection was generated by a computer and has not been reviewed by the radiologist for accuracy, correctness, or completeness. The illustration should NOT be used in medical decision making or diagnostic purposes. The illustration is provided solely as a convenience for the patient and should not be relied upon as a medical diagnosis. If you have questions relating to this illustration please contact your personal physician. The illustration is not intended to be a medical device and is not subject to FDA oversight.

Simplified Patient Explanation of Radiologist's Report*

Not a substitute for the Expert Radiology radiologist's report

MRI of the Lumbar Spine

Simplified Patient Explanation
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Clinical Indication
Low back pain after slipping and falling on 02/11/2026.
Technique
T1 and T2 images were taken.
Comparison
No previous studies to compare.
Findings
The lower part of the spinal cord looks normal. The curve of the lower spine is normal. No unusual lumps or fluid collections are seen. No unusual bone marrow changes are found.
Segmental Analysis

- **T12-L1:** There is a central disc herniation and a disc bulge together measuring 2.5 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There are bone spurs in the front and back. No narrowing of the spinal canal or nerve openings.
- **L1-2:** There is a disc bulge measuring 2.5 mm and a slight backward slip of L1 on L2 measuring 2 mm. The front part of the spinal sac is indented, and a ligament is lifted. There are bone spurs in the front and back. No narrowing of the spinal canal or nerve openings.
- **L2-3:** There is a central disc herniation and a disc bulge together measuring 2.5 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There are bone spurs in the front and back. No narrowing of the spinal canal or nerve openings.
- **L3-4:** There is a right side disc herniation and a disc bulge together measuring 3 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There is a slight backward slip of L3 on L4 measuring 2 mm. No narrowing of the spinal canal or nerve openings.

Simplified Patient Explanation
Simplified Patient Explanation

- **L4-5:** There is a left side disc herniation measuring 3.5 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There are bone spurs in the front and back. There is mild narrowing of the left nerve opening. No narrowing of the right nerve opening or spinal canal.
- **L5-S1:** There is a central disc herniation and a disc bulge together measuring 4 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There is moderate narrowing of the right nerve opening. No narrowing of the left nerve opening or spinal canal.

There is mild spinal asymmetry at L4. There is decreased disc signal from T12-L1 through L5-S1. There is decreased disc space height at L1-2. There are hemangiomas in T12, L2, and L5. There are Schmorl's nodes in the lower thoracic and lumbar spine. There is mild anterior wedging of the L1 vertebral body. There is increased signal in the ligament at L5-S1 showing injury or swelling. The prostate gland is enlarged.
Impression

- **T12-L1:** Central disc herniation and disc bulge together measuring 2.5 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There are bone spurs in the front and back.
- **L1-2:** Disc bulge measuring 2.5 mm and slight backward slip of L1 on L2 measuring 2 mm. The front part of the spinal sac is indented, and a ligament is lifted.
- **L2-3:** Central disc herniation and disc bulge together measuring 2.5 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There are bone spurs in the front and back.
- **L3-4:** Right side disc herniation and disc bulge together measuring 3 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. Slight backward slip of L3 on L4 measuring 2 mm.
- **L4-5:** Left side disc herniation measuring 3.5 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. There are bone spurs in the front and back. Mild narrowing of the left nerve opening.
- **L5-S1:** Central disc herniation and disc bulge together measuring 4 mm. The front part of the spinal sac is indented, and a ligament is lifted. There is a bright spot in the disc showing a tear. Moderate narrowing of the right nerve opening.

Mild spinal asymmetry at L4. Mild anterior wedging of the L1 vertebral body. Increased signal in the ligament at L5-S1 showing injury or swelling.

Simplified Patient Explanation

Simplified Patient Explanation

This MRI was done because of symptoms after a fall. The findings should be checked with the patient's doctors to see if they are related to the fall on 02/11/2026. This report should help decide the best treatment.Simplified Patient Explanation Simplified Patient Explanation

END OF SIMPLIFIED PATIENT EXPLANATION

**The above is a Simplified Patient Explanation / Artificial Intelligence ("AI") summary generated for the patient (a layperson) of the Radiology Report provided above ("Simplified Patient Explanation"). This Simplified Patient Explanation was generated by a computer and has not been reviewed by the radiologist for accuracy, correctness, or completeness. The Simplified Patient Explanation is not intended for decision making or diagnostic purposes. The Simplified Patient Explanation is provided solely as a convenience for the patient and should not be relied upon as a medical diagnosis. If you have questions relating to this AI Summary please contact your personal physician. The Simplified Patient Explanation by Expert Radiology™ is not intended to be a medical device and is not subject to FDA oversight.*

- Quantitative Evaluation of Large Language Models to Streamline Radiology Report Impressions: A Multimodal Retrospective Analysis. Radiology. https://doi.org/10.1148/radiol.231593
- Rahsepar, A. A. (2024). Large Language Models for Enhancing Radiology Report Impressions: Improve Readability While Decreasing Burnout. Radiology. https://doi.org/10.1148/radiol.240498
- Park, J., Oh, K., Han, K. et al. (2024). Patient-centered radiology reports with generative artificial intelligence: adding value to radiology reporting. https://doi.org/10.1038/s41598-024-63824-z

Physicians and healthcare providers only: For direct consultation with a radiologist at Expert Radiology™, please call our dedicated hotline at 415-900-2000.